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Multiple tuned mass damper control of vortex-induced vibration in bridges based on failure probability

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ABSTRACT

The multiple tuned mass damper (MTMD) is used instead of the single tuned mass damper (STMD) in this paper to improve the robustness in the mitigation of the vortex-induced vibration (VIV) in bridges. A framework of the parameter design of the MTMD mitigating the VIV is proposed based on a failure probability, which is derived as the objective function of the MTMD control by considering the uncertainties in bridge, aerodynamic, and MTMD parameters. Furthermore, a flowchart of the parameter optimization of the MTMD is presented by considering the failure probability and stroke limitation, simultaneously. At last, a numerical example of a box girder continuous bridge equipped with the optimized MTMD under the VIV is presented. Results show that the uncertainties in aerodynamic parameters characterizing the occurring condition and intensity of the VIV have great influence on the negative damping ratio induced by the VIV, while the uncertainties in natural frequencies of the bridge and MTMD have great influence on the equivalent damping ratio contributed by the MTMD. The superiority of the MTMD in the robustness will be lost if the intensity of the VIV is weak, but the MTMD is more robust than the STMD if the intensity of the VIV is modest or strong. The failure probability of the MTMD is 30 %∼50 % of that of the STMD when the total mass ratio is 0.6 %.

1. Introduction

The vortex-induced vibration (VIV) induced by the fluid–structure interaction occurs in long-span bridges at modest wind velocities [1,2]. Within the lock-in region of the VIV, the bridge girder generates a vortex-induced resonance with characteristics of large amplitude, long duration, and single mode [1–3]. The Rio-Niterói Bridge, Great Belt East Bridge, trans-Tokyo Bay Bridge, Xihoumen Bridge, and Humen Bridge had experienced the VIVs at modest wind velocities, with 250 mm amplitude [4], 350 mm amplitude and 100-min duration [5], 540 mm amplitude [6], 240 mm amplitude and 210-min duration [7], 200 mm amplitude and more than 120-min duration [8], respectively. Although the vortex-induced resonance is not catastrophic as the flutter, it causes the public panic and fatigue problem of structural components [1,2,7].

In order to mitigate these VIVs, the tuned mass dampers (TMDs) were installed in the girder to improve the damping ratio of the bridge, for example, the Great Belt East Bridge [9], Rio-Niterói Bridge [4], trans-Tokyo Bay Bridge [6], Judo Bridge [10], Chong Qi Bridge [11], and Hong Kong Zhuhai Macao Bridge[12]. However, there are some challenges in the TMD control of the VIV in bridges, they are the static stretching induced by gravity, auxiliary mass stroke, and robustness problem. Since the TMD is tuned to the natural frequency of flexible bridges, the static stretching and auxiliary mass stroke are large for the limited installation space or motion space. The major-minor frame design [6] and applying pre-stress design [11] were adopted to reduce the static stretching. Recently, the tuned mass damper inerter (TMDI) was used to simultaneously reduce the static stretching and auxiliary mass stroke [13–15]. The pounding viscoelastic TMD was used to reduce the auxiliary mass stroke, and its control effectiveness was validated by the sectional wind tuned test [16]. The robustness problem is significant for the single tuned mass damper (STMD) control due to the small mass ratio used in bridges [12], parameter estimation errors [17], dynamic characteristics fluctuation [18,19], and VIV uncertainty [20]. In order to improve the robustness, the magnetorheological TMD [20], active TMD [21], and bistable TMD [22] were proposed against uncertainties in bridge parameters and vortex-induced force. Wang et al. designed a

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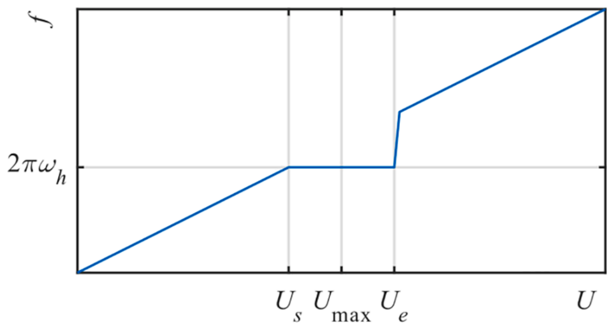


Fig. 1. The frequency of vortex shedding with the wind velocity.

novel independent variable mass TMD to improve the robustness in the mitigation of the walking-induced vibration in bridges [23], and this semi-active TMD was also promising in the mitigation of the VIV in bridges. In addition, optimizing the STMD parameters as a common strategy was widely adopted to weaken the mistuning sensitivity of the STMD for the VIV mitigation [6,12,24–26], but the improvement in the robustness was limited.

The multiple tuned mass damper (MTMD) consisting of some STMDs with non-uniform design parameters is more robust than the STMD [27]. Using the MTMD to suppress various structural vibrations had been extensively studied nearly-three decades, such as the harmonically forced oscillation [28], ground acceleration [29], seismic excitation [30], bridge flutter [31], human-induced vibration [32], wind turbine multiple hazards [33], and so on. The results showed that the effectiveness of the MTMD was slightly better than that of the STMD, while the robustness of the MTMD was much better than that of the STMD when MTMD parameters are reasonably optimized [27–33]. However, there are few studies on using the MTMD to mitigate the VIV in bridges. The MTMD may be more preferable than semi-active TMDs, active TMDs, and STMDs in practice by considering the configuration simplicity and control efficiency, simultaneously. The objective function of parameter optimization plays an important role in realizing a good MTMD control. The common objective functions are the norms of the transfer functions, peak structural response, root mean square of the structural response, minimum flutter velocity, etc [28–33]. Since the VIV in bridges within in the lock-in region is a self-excited vibration rather than a pure harmonic vibration, the amplitude of the girder will be completely eliminated once the equivalent damping ratio contributed by the STMD or MTMD exceeds a threshold. As a result, the amplitude of the girder cannot be used to evaluate the STMD or MTMD control of the nonlinear VIV in bridges with uncertainty. Li et al. [34] and Zhou et al. [35] proposed a failure probability for the vortex-induced amplitude of the girder based on a linear semi-empirical VIV model by considering the uncertainties in bridge and aerodynamic parameters. Dai et al. [26] proposed a failure probability for the VIV in bridges equipped with the STMD based on a nonlinear semi-empirical VIV model by considering the uncertainties in bridge, aerodynamic and STMD parameters, and compared the control effects of the three types of STMDs based on the proposed failure probability.

In this paper, a framework of the parameter design of the MTMD mitigating the VIV in bridges is proposed. The failure probability for the VIV in bridges equipped with the MTMD is proposed as the objective function of the MTMD control by considering uncertainties in bridge, aerodynamic, and MTMD parameters. The flowchart of the parameter optimization of the MTMD is presented by simultaneously considering the failure probability and auxiliary mass stroke. A numerical example of a box girder continuous bridge under the VIV, including the uncertainty analysis, failure probability analysis, and comparison analysis, is presented to validate better robustness of the MTMD control than the STMD control.

2. Failure probability for VIV in bridge

The equation of motion of unit length of a bridge subjected to the VIV is expressed by,

$$\ddot{h} + 2\xi_h\omega_h\dot{h} + \omega_h{}^2h = \frac{1}{m}\rho U^2 D\left(Y_1(K)\left(1-\varepsilon\frac{h^2}{D^2}\right)\frac{\dot{h}}{U} + Y_2(K)\frac{h}{D} + \frac{1}{2}C_{LV}\sin(\omega t+\theta)\right)\# \tag{1}$$

where h is the vertical amplitude of unit length; m, ω_h, and ξ_h are the mass, natural frequency, and damping ratio of unit length; ρ is the air density; U is the mean wind velocity; D is the height of the cross-section; $K = \omega D/U$ is the reduced frequency of vortex shedding; ω is the circular frequency of vortex shedding; Y_1, ε, Y_2, and C_{LV} are parameters identified by wind tunnel tests; θ is the phase angle.

Within the lock-in region, the VIV is dominated by the self-excited vibration, that is to say, the second and third items of the right of Eq. (1) can be dropped [1,2,24,36]. Therefore, Eq. (1) at lock-in can be simplified by,

$$\ddot{h} + 2\xi_h\omega_h\dot{h} + \omega_h{}^2h = \frac{1}{m}\rho U^2 DY_1\left(1-\varepsilon\frac{h^2}{D^2}\right)\frac{\dot{h}}{U} \tag{2}$$

Introducing $h(x) = \phi(x)q$, Eq. (2) is written as the equation of motion in modal coordinates,

$$\ddot{q}+\left(2\xi_h\omega_h-\frac{\rho UDY_1}{M}\int_0^L\phi^2(x)\mathrm{d}x+\frac{\varepsilon\rho UY_1}{MD}\int_0^L\phi^2(x)\phi^2(x)\mathrm{d}xq^2\right)\dot{q}+\omega_h{}^2q=0\# \tag{3}$$

where ϕ is the mode shape; q is the generalized modal coordinate; $M = \int_0^L\phi^2(x)m(x)\mathrm{d}x$ is the generalized mass of the bridge; L is the length of the bridge.

Introducing $\alpha_2 = \int_0^L\phi^2(x)\mathrm{d}x$ and $\alpha_4 = \int_0^L\phi^4(x)\mathrm{d}x$, Eq. (3) is rewritten as,

$$\ddot{q}+\left(2\xi_h\omega_h-\frac{\alpha_2\rho UDY_1}{M}+\frac{\alpha_4\varepsilon\rho UY_1}{MD}q^2\right)\dot{q}+\omega_h{}^2q=0 \tag{4}$$

Noted that, the form of Eq. (4) is similar to that of a classical van der Pol oscillator [36],

$$\ddot{x}+\epsilon\left(x^2-1\right)\dot{x}+x=0 \tag{5}$$

where x is the amplitude of the van der Pol oscillator. For small value of ϵ, the nonlinearity of the van der Pol oscillator is weak, and its motion is nearly sinusoidal. Experimental and filed test results showed that the nonlinearity in the VIV in bridges is weak, thus the Krylov-Bogoliubov method (KBM) for the van der Pol oscillator with small ϵ can be also used to solve Eq. (4),

$$q_0 = 2\sqrt{\frac{\alpha_2 D^2}{\alpha_4 \varepsilon} - \frac{2\xi_h \omega_h MD}{\alpha_4 \varepsilon \rho U Y_1}} \tag{6}$$

where q_0 is the approximate steady-state solution of q.

Assuming $M = m\alpha_2$, and introducing the Strouhal number $S_t = \omega_h D/(2\pi U)$ and the mass ratio $m_r = \rho D^2/m$, the dimensionless steady-state amplitude of the bridge at lock-in with $\phi(x_{\max}) = 1$ is written as,

$$\frac{h_0}{D} = 2\sqrt{\frac{\alpha_2}{\varepsilon \alpha_4}}\sqrt{1 - \frac{4\pi \xi_h S_t}{m_r Y_1}} \tag{7}$$

where h_0 is the approximate steady-state solution of h; $x_{\max}$ is the maximum coordinate along the length of the bridge.

As noted in Fig. 1, the lock-in region is located between the start wind velocity U_s and end wind velocity U_e, accordingly the Strouhal number has an interval, rather than a constant. In this study, we assume that the maximum steady-state amplitude occurs at the wind velocity $U_{\max} = (U_s + U_e)/2$, then the Strouhal number is rewritten as,

$$S_t(U_{\max}) = \frac{2}{(1 + L_l)} S_{t,s} \tag{8}$$

where $S_{t,s} = \omega_h D/(2\pi U_s)$; $L_l = U_e/U_s$ is defined as lock-in region parameter [34].

Substituting Eq. (8) into Eq. (7) yields,

$$\frac{h_0}{D} = 2\sqrt{\frac{\alpha_2}{\varepsilon \alpha_4}}\sqrt{1 - \frac{8\pi \xi_h S_{t,s}}{(1 + L_l) m_r Y_1}} \# \tag{9}$$

According to Eq. (9), aerodynamic parameters Y_1 and ε represent the magnitude of the vortex-induced amplitude. ε regulates the maximum amplitude allowed at the limit $S_t \to 0$, Y_1 characterize the energy feeding into the system and it is crucial for the design of the TMD [36].

Introducing the maximum allowable vortex-induced amplitude of the bridge $[h]$ given by the standard or specification, the limit state function Z_{I} for judging whether the VIV needs to be mitigated is obtained,

$$Z_{\mathrm{I}} = \frac{[h]}{D} - 2\sqrt{\frac{\alpha_2}{\varepsilon \alpha_4}}\sqrt{1 - \frac{8\pi \xi_h S_{t,s}}{(1 + L_l) m_r Y_1}} \tag{10}$$

According to Eq. (10), the VIV needs be mitigated if $h_0 > [h]$ ($Z_{\mathrm{I}} < 0$), and the VIV does not need to be mitigated if $h_0 \le [h]$ ($Z_{\mathrm{I}} \ge 0$). Finally, the failure probability for the VIV in bridges can be obtained,

$$P_{f_{\mathrm{I}}} = P\{Z_{\mathrm{I}} < 0\} \tag{11}$$

where $P_{f_{\mathrm{I}}}$ is the failure probability for the VIV in the bridge.

3. Failure probability for VIV in bridge with MTMD

In wind tunnel tests, the amplitude under various damping ratio is measured to evaluate the VIV risk of the bridge. Thus, the damping ratio instead of the amplitude here is adopted to judge whether the VIV needs to be mitigated, $Z_{\mathrm{I}} < 0$ is rewritten as,

$$\xi_h < \frac{(1 + L_l) m_r Y_1}{8\pi S_{t,s}} \left(1 - \frac{4\varepsilon \alpha_4 [h]^2}{\alpha_2 D^2}\right) \tag{12}$$

If the contribution of the MTMD to the VIV mitigation is also expressed by the damping ratio, a new limit state function Z_{II} for judging whether the MTMD control meets the VIV mitigation goal is proposed,

$$Z_{\mathrm{II}} = (\xi_e + \xi_h) - \frac{(1 + L_l) m_r Y_1}{8\pi S_{t,s}} \left(1 - \frac{\varepsilon \alpha_4 [h]^2}{4\alpha_2 D^2}\right) \tag{13}$$

where ξ_e is the equivalent damping ratio provided by the MTMD, the right item of Eq. (12) is defined as the negative damping ratio of the bridge, which is induced by the vortex shedding in the VIV.

In order to obtain the expression of ξ_e, the equations of motion of the bridge with the MTMD in modal coordinates are firstly given by,

$$\ddot{q} + \left(2\xi_h \omega_h - \frac{\alpha_2 \rho U D Y_1}{M} + \frac{\alpha_4 \varepsilon \rho U Y_1}{MD} q^2\right)\dot{q} + \omega_h^2 q = \sum_{n=1}^{N} \mu_n \left[\omega_{t,n}^2 \left(q_{t,n}(t) - q(t)\right) + 2\xi_{t,n}\omega_{t,n}\left(\dot{q}_{t,n}(t) - \dot{q}(t)\right)\right] \tag{14a}$$

$$\ddot{q}_{t,n} + \omega_{t,n}^2 (q_{t,n} - q) + 2\xi_{t,n}\omega_{t,n}\left(\dot{q}_{t,n} - \dot{q}\right) = 0 \tag{14b}$$

where $q_{t,n}$ is the generalized modal coordinate of the n th TMD; $\mu_n = m_n/M$, $\omega_{t,n} = \sqrt{k_n/m_n}$ and $\xi_{t,n} = \sqrt{c_n/(2m_n\omega_{t,n})}$ are the mass ratio, natural frequency, and damping ratio of the n th TMD; m_n, k_n, and c_n are the mass, stiffness, and damping of the n th TMD; N is the number of the TMD.

Accroding to Eq. (14b), the relationship between q and $q_{t,n}$ is expressed as,

$$q_{t,n} = \frac{\omega_{t,n}^2 + 2\xi_{t,n}\omega_{t,n}\omega i}{-\omega^2 + \omega_{t,n}^2 + 2\xi_{t,n}\omega_{t,n}\omega i} q \tag{15}$$

The normalized force applied to the bridge by the n th TMD can be obtained after substituting Eq. (15) into the right of Eq. (14a),

$$F_n = \mu_n \left[\omega_{t,n}^2 (q_{t,n} - q) + 2\xi_{t,n}\omega_{t,n}\left(\dot{q}_{t,n} - \dot{q}\right)\right] = \mu_n \omega^2 \frac{\omega_{t,n}^2 + 2\xi_{t,n}\omega_{t,n}\omega i}{-\omega^2 + \omega_{t,n}^2 + 2\xi_{t,n}\omega_{t,n}\omega i} q \# \tag{16}$$

Introducing $\beta_n = \omega_{t,n}/\omega_h$ and $\gamma = \omega/\omega_h$, Eq. (16) is rewritten as,

$$F_n = \mu_n \omega^2 \frac{-\beta_n^2\gamma^2 + \beta_n^4 + 4\xi_{t,n}^2\beta_n^2\gamma^2 - 2\xi_{t,n}\beta_n\gamma^3 i}{\left(-\gamma^2 + \beta_n^2\right)^2 + 4\xi_{t,n}^2\beta_n^2\gamma^2} q = \mu_n \omega_h^2 \gamma^2 \frac{a_n}{c_n} q + \mu_n \omega_h \frac{b_n}{c_n} \gamma \dot{q} \# \tag{17}$$

where $a_n = -\beta_n^2\gamma^2 + \beta_n^4 + 4\xi_{t,n}^2\beta_n^2\gamma^2$, $b_n = -2\xi_{t,n}\beta_n\gamma^3$, and $c_n = \left(-\gamma^2 + \beta_n^2\right)^2 + 4\xi_{t,n}^2\beta_n^2\gamma^2$.

Substituting Eq. (17) into Eq. (14a) yields,

$$\ddot{q} + \omega_h^2 q = f(\dot{q}, q) \tag{18}$$

$$f(\dot{q}, q) = -\frac{\alpha_4 \varepsilon \rho U Y_1}{MD} q^2 \dot{q} + \left(-2\xi_h\omega_h + \frac{\alpha_2 \rho U D Y_1}{M} + \omega_h \sum_{n=1}^{N} \mu_n \frac{\gamma b_n}{c_n}\right)\dot{q} + \left(\omega_h^2\gamma^2 \sum_{n=1}^{N} \mu_n \frac{a_n}{c_n}\right) q \# \tag{19}$$

where $\alpha_4 \varepsilon \rho U Y_1 q^2 \dot{q}/MD$ is the weak nonlinear item of the bridge-MTMD system within the lock-in region. According to the KBM, q and $\dot{q}$ are assumed as,

$$q = a(t)cos(\omega t + \varphi(t)) \tag{20}$$

$$\dot{q} = -a(t)\omega \sin(\omega t + \varphi(t)) \tag{21}$$

Table 1
The probability distribution of random variables.

Variable	Distribution	Reference	Variable	Distribution	Reference
$S_{t,s}$	Log-normal	[37]	α_2	Normal	Assumed
L_l	Extreme value I	[34,37]	α_4	Normal	Assumed
Y_1	Normal	Assumed	ω_h	Log-normal	[37]
ε	Normal	Assumed	ξ_h	Log-normal	[37]
D	Normal	[37]	$\omega_{t,n}$	Log-normal	[37]
m	Normal	[37]	$\xi_{t,n}$	Log-normal	y[37]

where $a(t)$ and $\varphi(t)$ are functions of time rather than constants in this nonlinear system. Noted that, the establishment of the relationship between Eqs. (20) and (21) should satisfies the following equation,

$$\dot{a}cos(\omega t+\varphi)-a\dot{\varphi}sin(\omega t+\varphi)=0 \tag{22}$$

Taking the derivative of $\dot{q}$ in Eq. (21), $\ddot{q}$ can be expressed as,

$$\ddot{q}=-\dot{a}\omega\sin\psi-a\omega^2\cos\psi-a\omega\dot{\varphi}\cos\psi \tag{23}$$

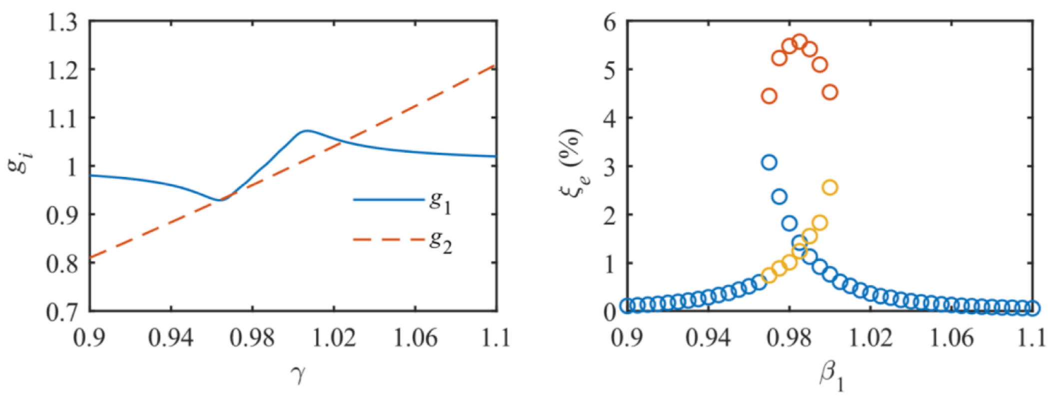
where $\psi=\omega t+\varphi$. Substituting Eq. (23) into Eq. (18) yields,

$$-\dot{a}\omega\sin\psi-a\omega\dot{\varphi}\cos\psi=f(\dot{q},\ q)+a\left(\omega^2-{\omega_h}^2\right)\cos\psi \tag{24}$$

According to Eqs. (22) and (24), the expressions of $\dot{a}$ and $\dot{\varphi}$ are solved,

$$\dot{a}=-\frac{1}{\omega}\left[f(\dot{q},\ q)+a\left(\omega^2-{\omega_h}^2\right)\cos\psi\right]\sin\psi \tag{25a}$$

$$\dot{\varphi}=-\frac{1}{a\omega}\left[f(\dot{q},\ q)+a\left(\omega^2-{\omega_h}^2\right)\cos\psi\right]\cos\psi \tag{25b}$$



(a) The multiple resonance frequency ratios (b) The multiple equivalent damping ratios

Fig. 2. The multiple solutions of γ and ξ_e in the LFD MTMD control.

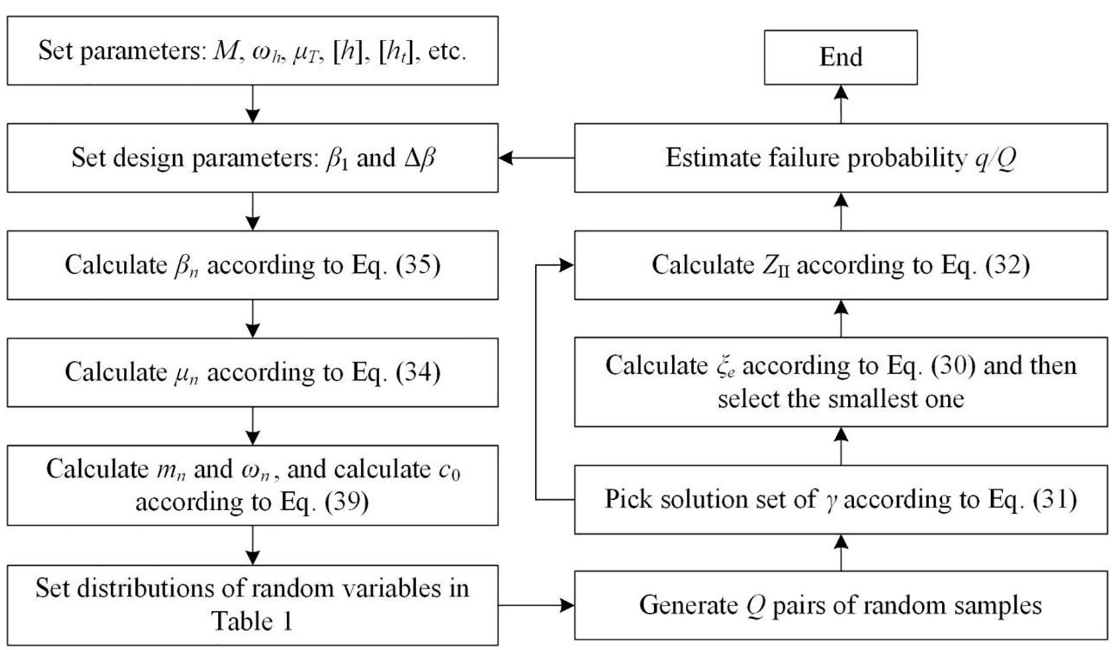


Fig. 3. The flowchart of the parameter optimization of the MTMD.

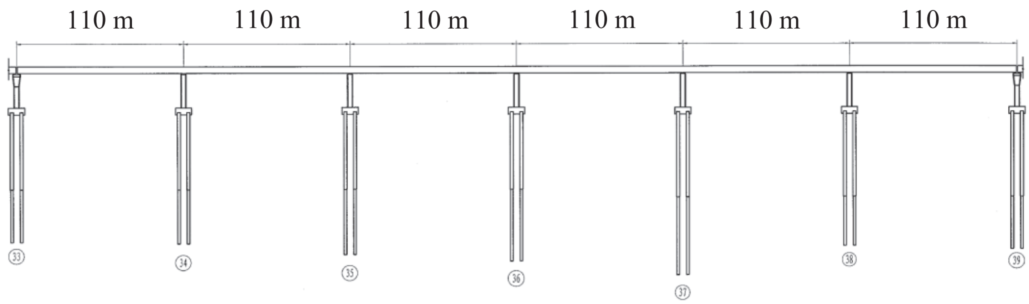


Fig. 4. The 6 × 110 m box girder continuous bridge.

Table 2
The bridge parameters and aerodynamic parameters.

Parameter	D	M	ω_h	ξ_h	α_2	α_4
Value	4.5 m	8911.4 ton	5.064 rad/s	0.3 %	29.98	22.469
Parameter	U_s	U_e	$U_{\max}$	Y_1	ε	
Value	27 m/s	33 m/s	30 m/s	179.108	2561.843	

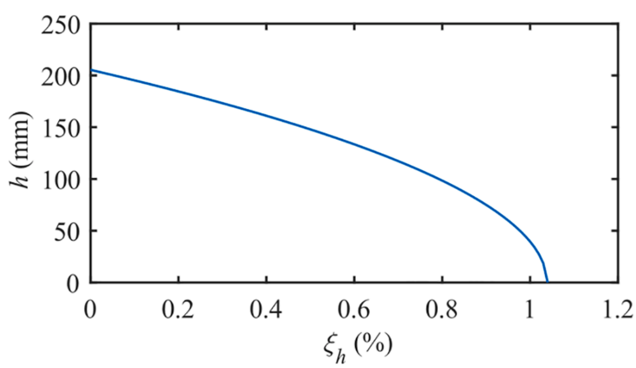


Fig. 5. The relationship between the resonant amplitude and damping ratio of the bridge.

Table 3
The COV of the random variables in the negative damping ratio.

Variable	$S_{t,s}$	L_l	Y_1	ε	D	m	α_2	α_4	ω_h
COV (%)	0 ~ 15	0 ~ 15	0 ~ 15	0 ~ 15	0 ~ 5	0 ~ 5	0 ~ 10	0 ~ 10	0 ~ 10

Since $a(t)$ and $\varphi(t)$ both vary slowly with time, the right items of Eqs. (25) can be approximately replaced by their averages during a period,

$$\dot{a} = -\frac{1}{2\pi\omega}\int_0^{2\pi}\left[f(\dot{q},\ q) + a\left(\omega^2 - {\omega_h}^2\right)\cos\psi\right]\sin\psi d\psi \tag{26a}$$

$$\dot{\varphi} = -\frac{1}{2\pi a\omega}\int_0^{2\pi}\left[f(\dot{q},\ q) + a\left(\omega^2 - {\omega_h}^2\right)\cos\psi\right]\cos\psi d\psi \tag{26b}$$

Here, we only consider the steady-state solution of a and φ, thus Eqs. (26) can be rewritten as,

$$\frac{1}{2\pi\omega}\int_0^{2\pi}\left[f(\dot{q},\ q) + a\left(\omega^2 - {\omega_h}^2\right)\cos\psi\right]\sin\psi d\psi = 0 \tag{27a}$$

$$\frac{1}{2\pi a\omega}\int_0^{2\pi}\left[f(\dot{q},\ q) + a\left(\omega^2 - {\omega_h}^2\right)\cos\psi\right]\cos\psi d\psi = 0 \tag{27b}$$

Substituting Eqs. (19), (20), and (21) into Eqs. (27) and then solving Eqs. (27), two algebraic equations are obtained,

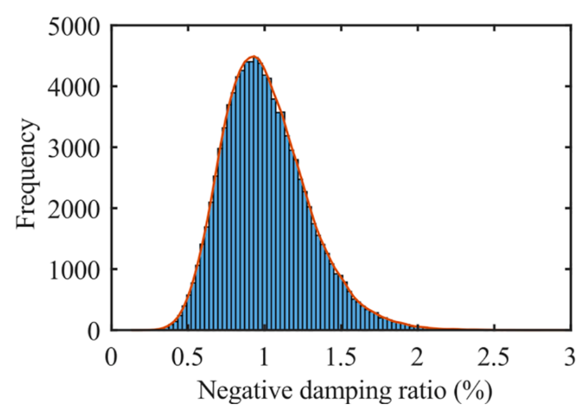


Fig. 7. The frequency histogram of the negative damping ratio.

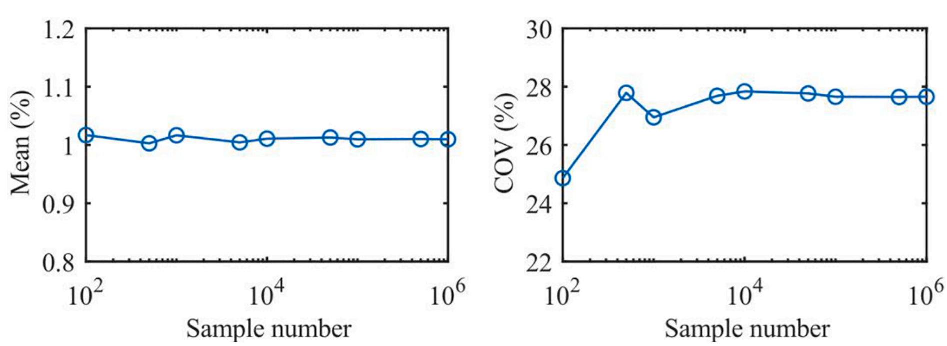


Fig. 6. The mean and COV of the negative damping ratio varying with the sample number.

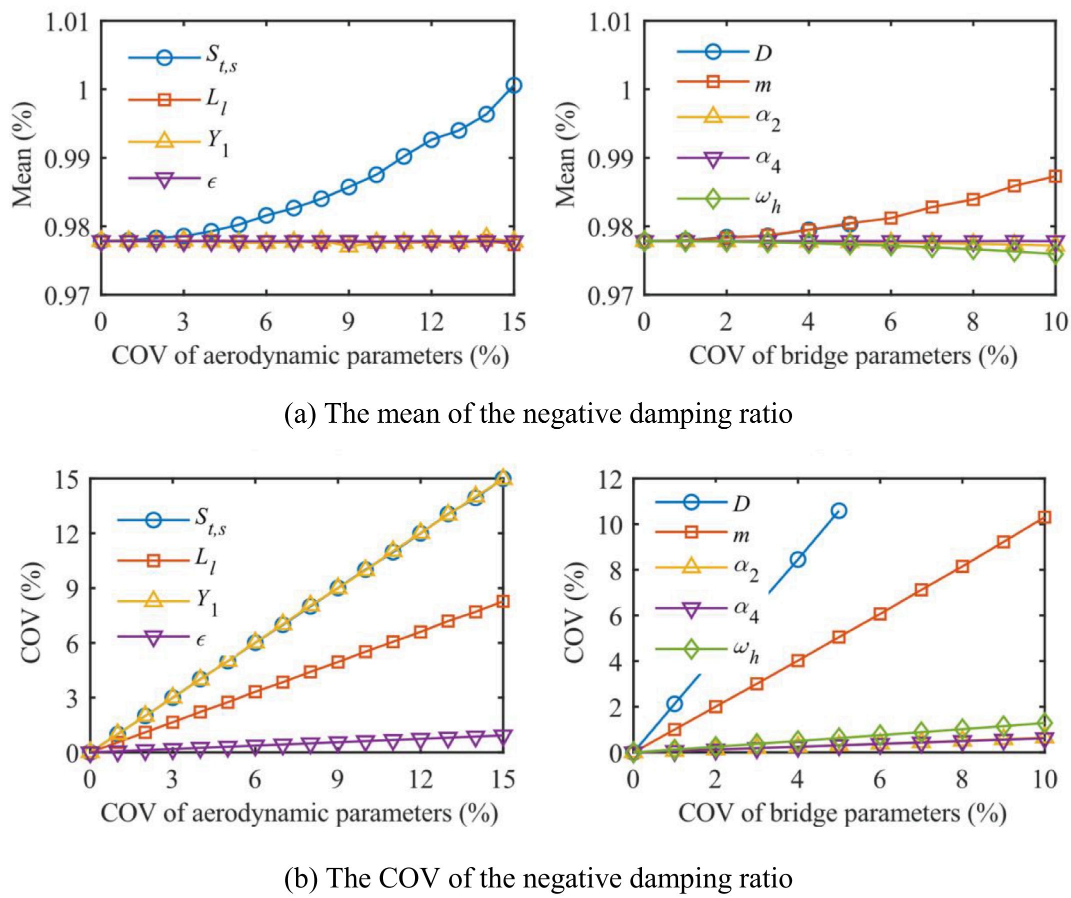


Fig. 8. The mean and COV of the negative damping ratio varying with the COV of each random variable.

Table 4
The parameters of LMD and LFD MTMDs.

	Number	μ_T	β_1	$\Delta\beta$	ξ_t
LMD	4	0.2 %, 0.4 %, 0.6 %	1	$\Delta\beta_M = 10$	5 %, 7 %, 10 %
LFD			0.97	$\Delta\beta_F = 0.01$	

$$\frac{\alpha_4 \varepsilon \rho U Y_1}{4MD} a^2 + 2\xi_h \omega_h - \frac{\alpha_2 \rho U D Y_1}{M} - \omega_h \sum_{n=1}^{N} \mu_n \frac{\gamma b_n}{c_n} = 0 \tag{28a}$$

$$\gamma^2 \sum_{n=1}^{N} \mu_n \frac{a_n}{c_n} + \left(\gamma^2 - 1\right) = 0 \tag{28b}$$

The steady-state amplitude of the bridge can be obtained by solving Eq. (28a),

$$q_0 = a = 2\sqrt{\frac{\alpha_2 D^2}{\alpha_4 \varepsilon} - \frac{2MD\left(\xi_h - \frac{1}{2}\sum_{n=1}^{N}\mu_n \frac{\gamma b_n}{c_n}\right)\omega_h}{\alpha_4 \varepsilon \rho U Y_1}} \tag{29}$$

By comparing Eq. (6) and Eq. (29), we can find that the role of the MTMD in the VIV mitigation is equivalent to increasing the damping ratio of the bridge, thus the equivalent damping ratio in Eq. (13) is expressed as,

$$\xi_e = -\frac{1}{2}\sum_{n=1}^{N} \mu_n \frac{\gamma b_n}{c_n} \tag{30}$$

where γ is the vortex-induced resonance frequency ratio, which is obtained by solving Eq. (28b). Since the analytical solution of Eq. (28b) cannot be obtained, the intersections of two curves here are picked as solution set of γ. The functions of the two curves are,

Table 5
The comparison result of the equivalent damping ratio.

	μ_T	$\xi_t = 5\%$		$\xi_t = 7\%$		$\xi_t = 10\%$	
LMD	0.2 %	0.959 %#	0.923 %*	0.701 %# 39.01 mm#	38.31 mm*	0.496 %# 99.25 mm#	99.33 mm*
	0.4 %	1.909 %#	2.485 %*	1.403 %#	1.453 %*	0.992 %#	0.971 %*
	0.6 %	2.847 %#	3.222 %*	2.106 %#	2.734 %*	1.486 %#	1.533 %*
LFD	0.2 %	0.863 %#	0.837 %*	0.662 %# 55.73 mm#	54.15 mm*	0.482 %# 102.1 mm#	101.6 mm*
	0.4 %	1.678 %#	1.849 %*	1.325 %#	1.35 %*	0.965 %#	0.946 %*
	0.6 %	2.438 %#	2.632 %*	1.979 %#	2.316 %*	1.447 %#	1.486 %*

Note: # and * denote the analytical result and time-domain simulation (true) result, respectively.

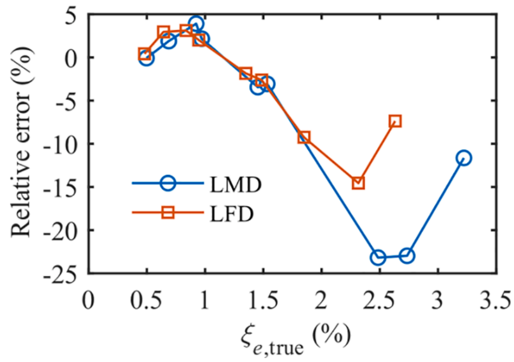


Fig. 9. The relative error between the approximate analytical result and true result.

$$g_1(\gamma) = 1 - \sum_{n=1}^{N} \mu_{1,n} \frac{\gamma^2 a_n}{c_n} \tag{31a}$$

$$g_2(\gamma) = \gamma^2 \tag{31b}$$

Substituting Eq. (30) into Eq. (13) yields,

$$Z_{\mathrm{II}} = \xi_h - \frac{1}{2}\sum_{n=1}^{N} \mu_n \frac{\gamma b_n}{c_n} - \frac{(1+L_l) m_r Y_1}{8\pi S_{t,s}} \left(1 - \frac{\varepsilon \alpha_4 [h]^2}{4\alpha_2 D^2}\right) \tag{32}$$

Finally, the failure probability for the VIV in the bridge with the MTMD is obtained,

$$P_{f_{\mathrm{II}}} = P\{Z_{\mathrm{II}} < 0\} \tag{33}$$

where $P_{f_{\mathrm{II}}}$ is the failure probability for the VIV in the bridge with the MTMD.

Since popular VIV mathematical models cannot accurately describe the behaviour of the VIV, at the same time, the control effect of the small mass ratio TMD is very sensitive to the parameter estimation errors and dynamic characteristics fluctuation, thus the uncertainty in the bridge-MTMD system within the lock-in region and its effect on the VIV mitigation cannot be neglected. Based on the two facts, main parameters in the limit state function Z_{II} are considered as random variables. Table 1 presents the probability distribution of these random variables. Although there are no statistics for the uncertainties in Y_1, ε, α_2, and α_4, we assume that they all obey the normal distribution, which is widely used in the natural science and social science to represent random variables whose distributions are not known. Such a normal distribution is also adopted in the wind engineering of bridges [34,35,38].

4. Parameter optimization of MTMD

If the equivalent damping ratio contributed by the MTMD is large and robust enough to uncertainties, the failure probability in Eq. (33) will be small. Therefore, the parameters of the MTMD are optimized here by using the failure probability as the performance function. In fact, it is challenging to realize the parameter optimization of a non-uniformly distributed MTMD if the uncertainties are introduced [39]. To simplify the parameter optimization and consider the practical implementation, the total mass ratio $\sum_{n=1}^{N}\mu_n = \mu_T$ is firstly given, the stiffness and damping coefficient of each TMD are identical, i.e., $k_n = k_0$ and $c_n = c_0$, and the MTMD is assumed to have linearly and uniformly distributed design parameters [40].

If $k_n = k_0$ is determined, the relationship between μ_n and μ_T can be obtained,

$$\mu_n = \frac{\frac{1}{\beta_n^2}}{\sum_{n=1}^{N}\frac{1}{\beta_n^2}} \mu_T \tag{34}$$

There are two types of linearly distributed parameters for the mass ratio determination, the first one is called the linear mass distribution

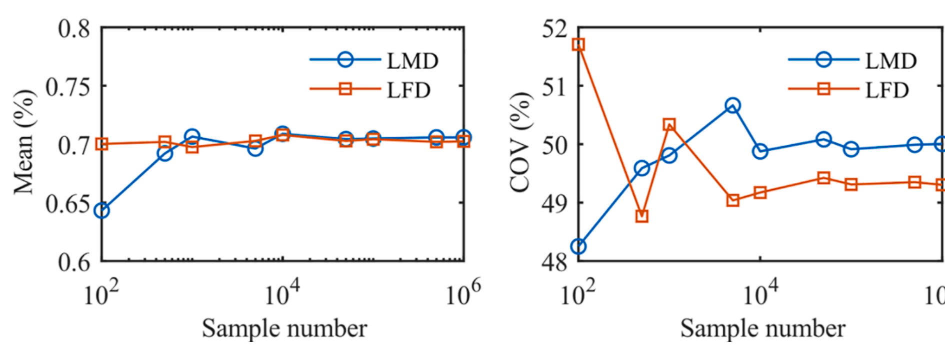


Fig. 10. The mean and COV of the equivalent damping ratio varying with the sample number.

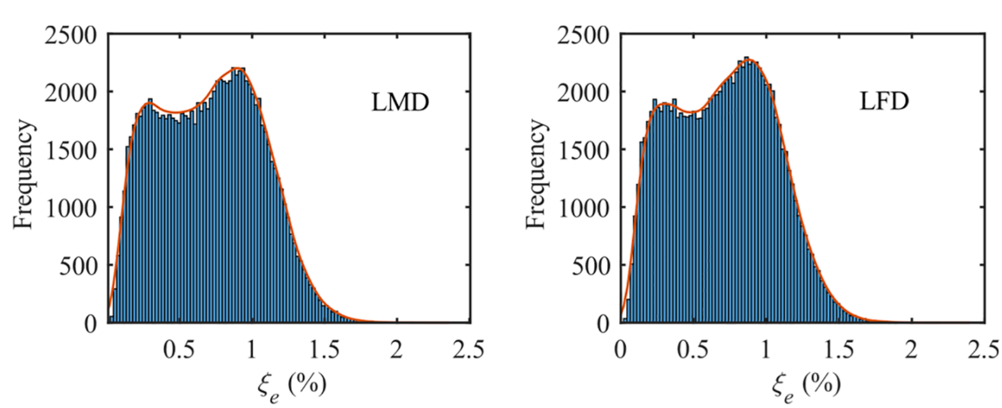
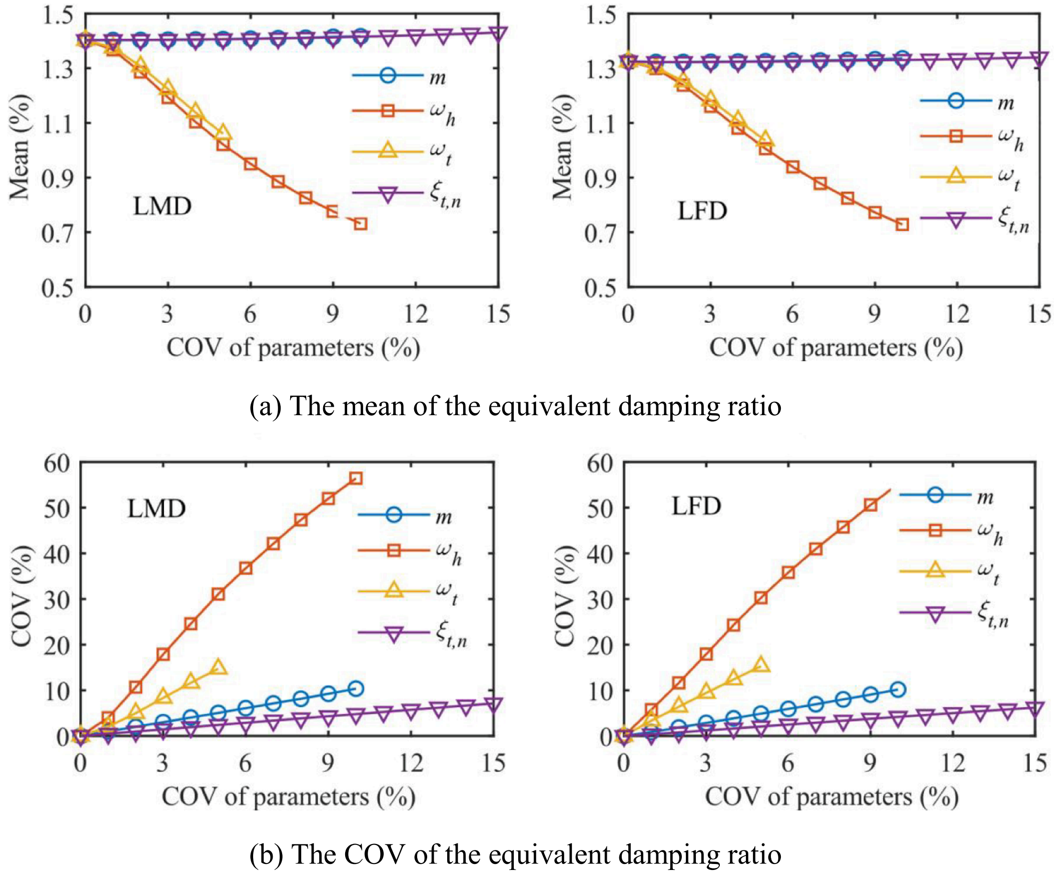


Fig. 11. The frequency histogram of the equivalent damping ratio.



(a) The mean of the equivalent damping ratio

(b) The COV of the equivalent damping ratio

Fig. 12. The mean and COV of the equivalent damping ratio varying with the COV of each random variable.

Table 7
The optimal parameters of the MTMDs and STMD.

Type	μ_T	$\xi_t = 5\%$	$\xi_t = 7\%$	$\xi_t = 10\%$
TMD	0.2 %	$\beta = 0.999$	$\beta = 1.006$	$\beta = 1.007$
	0.4 %	$\beta = 0.995$	$\beta = 0.998$	$\beta = 0.999$
	0.6 %	$\beta = 0.989$	$\beta = 0.987$	$\beta = 0.988$
LMD	0.2 %	$\beta_1 = 1$, $\Delta\beta_M = 22$	$\beta_1 = 1.01$, $\Delta\beta_M = 19$	$\beta_1 = 1.01$, $\Delta\beta_M = 29$
	0.4 %	$\beta_1 = 1.05$, $\Delta\beta_M = 4$	$\beta_1 = 1.05$, $\Delta\beta_M = 4$	$\beta_1 = 1$, $\Delta\beta_M = 15$
	0.6 %	$\beta_1 = 1.1$, $\Delta\beta_M = 3$	$\beta_1 = 1.1$, $\Delta\beta_M = 3$	$\beta_1 = 1.04$, $\Delta\beta_M = 4$
LFD	0.2 %	$\beta_1 = 0.999$, $\Delta\beta_F = 0$	$\beta_1 = 1.006$, $\Delta\beta_F = 0$	$\beta_1 = 1.007$, $\Delta\beta_F = 0$
	0.4 %	$\beta_1 = 0.938$, $\Delta\beta_F = 0.042$	$\beta_1 = 0.956$, $\Delta\beta_F = 0.028$	$\beta_1 = 0.999$, $\Delta\beta_F = 0$
	0.6 %	$\beta_1 = 0.91$, $\Delta\beta_F = 0.062$	$\beta_1 = 0.92$, $\Delta\beta_F = 0.054$	$\beta_1 = 0.94$, $\Delta\beta_F = 0.036$

(LMD) $m_n = m_1 + (n-1)\Delta m$, where Δm is the mass spacing of the MTMD, and the second one is called the linear frequency distribution (LFD) $\omega_n = \omega_1 + (n-1)\Delta\omega$, where $\Delta\omega$ is the frequency spacing of the MTMD.

Then, the frequency ratios under the two distributions are respectively expressed as,

$$\frac{1}{{\beta_n}^2} = \frac{1}{{\beta_1}^2} + (n-1)\frac{1}{{\Delta\beta_M}^2} \tag{35a}$$

$$\beta_n = \beta_1 + (n-1)\Delta\beta_F \tag{35b}$$

where $\Delta\beta_M$ and $\Delta\beta_F$ are the frequency ratio spacings of the LMD and LFD MTMDs, respectively. The damping ratio of the n th TMD is expressed as,

$$\xi_{t,n} = \frac{c_0}{2M\omega_h}\frac{1}{\mu_n\beta_n} \tag{36}$$

In the VIV mitigation of the bridge, the MTMD stroke should be seriously considered due to the limited height of the girder. Besides, in order to avoid a dual-objective optimization involving the failure probability and MTMD stroke, the damping coefficient of each TMD c_0 needs to be determined before the optimization. According to Eq. (14), we can obtain,

$$r = \left|\frac{q_{t,n} - q}{q}\right| = \left|\frac{\gamma^2}{-\gamma^2 + {\beta_n}^2 + 2\xi_{t,n}\beta_n\gamma i}\right| \approx \frac{1}{2\xi_{t,n}} \tag{37}$$

where r is the ratio of the MTMD stroke and amplitude of the bridge. According to Eq. (37), an extreme scenario for the MTMD stroke capability is constructed: (a) the MTMD stroke reaches its maximum value $[h_t]$, and (b) the amplitude of the bridge reaches its limit value $[h]$, then the damping ratio the MTMD is written as,

$$\xi_t = \frac{[h]}{2([h_t] - [h])} \tag{38}$$

Furthermore, we assume that the μ_n and β_n of each TMD have no influence on its stroke, and according to Eqs. (36) and (38), the identical damping coefficient is expressed as,

$$c_0 = M\mu_T\omega_h\frac{[h]}{([h_t] - [h])} \tag{39}$$

Based on the above, we find that there are only two parameters β_1 and $\Delta\beta_M$ (or $\Delta\beta_F$) for optimization when the total mass ratio μ_T, maximum MTMD stroke $[h_t]$, and $[h]$ are given. It is noted from Eq. (31)

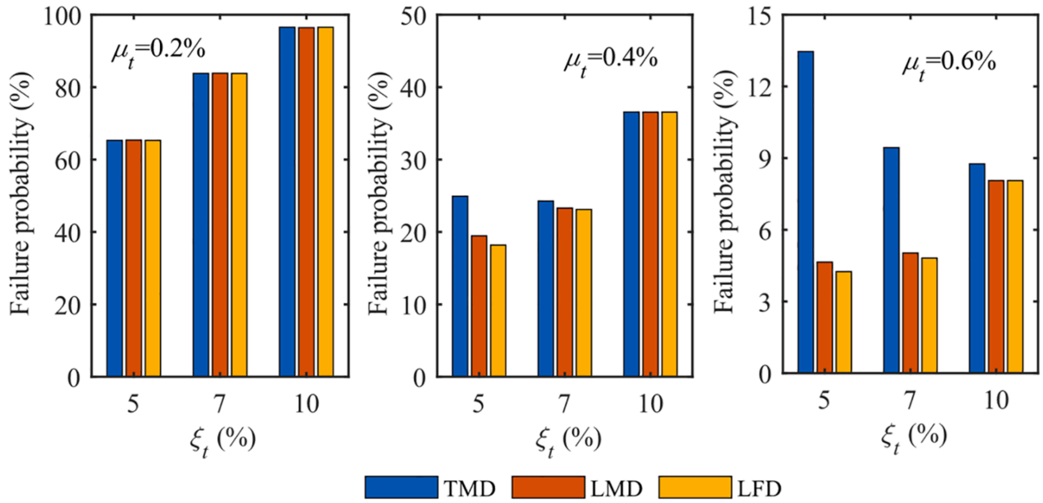


Fig. 13. The comparison result between the MTMD controls and STMD control.

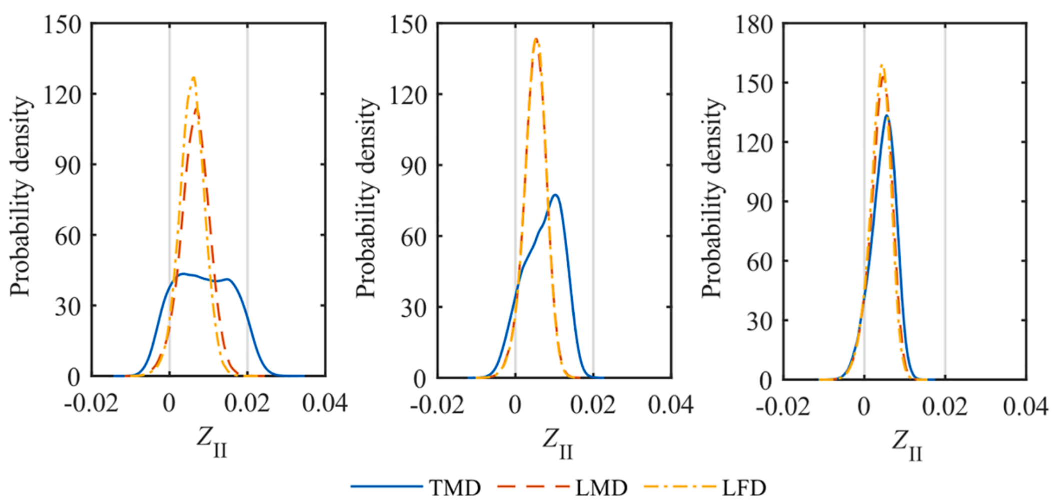


Fig. 14. The comparison result of the fitted probability distribution curve of the limit state function between the MTMDs and STMD.

that there will be a solution set rather than a solution for γ when $\beta_n \approx 1$ and ξ_t is small [24,26]. For example, Fig. 2(a) shows the multiple resonance frequency ratios in the LFD MTMD control. The design parameters of the LFD MTMD are $\mu_T = 0.4\%$, $N = 4$, $\xi_t = 2\%$, $\beta_1 = 0.97$, and $\Delta\beta_F = 0.01$. Accordingly, substituting the multiple resonance frequency ratios into Eq. (30) yields multiple equivalent damping ratios, they are 4.448 %, 3.079 %, and 0.745 %. Furthermore, the multiple equivalent damping ratios varying with β_1 are plotted in Fig. 2(b). Since the Eq. (14) is a weak nonlinear equation, a multiple-solution phenomenon (multiple equivalent damping ratios) was found when β_1 is close to 1, causing an ambiguous evaluation on the control effect of the MTMD [24,26]. In order to obtain a reliable control effect, the smallest one among multiple equivalent damping ratios is adopted for the failure probability analysis.

Since there are multiple solutions mentioned above, it is difficult to obtain the analytical failure probability. Instead, the Monte Carlo Simulation (MCS) with high accuracy is used to estimate the failure probability,

$$P_{f_{\mathrm{II}}} \approx \frac{q}{Q} \tag{40}$$

where Q is the number of the samples; q is the number of the result with $Z_{\mathrm{II}} < 0$. Fig. 3 shows the flowchart of the parameter optimization of the MTMD.

5. Numerical example

In order to validate the superiority of the MTMD control, a numerical example of a box girder continuous bridge subjected to the VIV is presented. The investigated bridge is a 6 × 110 m bridge with a height of 4.5 m and a width of 33.1 m, as shown in Fig. 4. The modal analysis of the bridge were conducted in our previous research [12].

5.1. The VIV in a box girder continuous bridge

The wind tunnel tests of the bridge were conducted by Qin et al., and the details of wind tunnel tests can be found in [41]. The results showed that a VIV with the first order vertical bending mode (the natural frequency f_h is 0.806 Hz) occurred at wind velocities between 27 m/s and 35 m/s, and the resonant amplitude reached 150 mm exceeding the limit value suggested in China code, $[h] = 40/f_h = 50$mm. Table 2 presents the bridge parameters and aerodynamic parameters. Fig. 5 shows the relationship between the resonant amplitude and damping ratio of the bridge. The resonant amplitude is 205 mm by neglecting the inherent damping ratio of the bridge, and it will be eliminated when the inherent damping ratio reaches 1.04 %, which indicates that the negative damping induced by vortex shedding is considerable. Since the full correlation of the aerodynamic forces along the span of the bridge is considered by adding the item of $\sqrt{\alpha_2/\alpha_4}$, the analytical resonant amplitude 173 mm is larger than 150 mm that tested in the sectional model. The resonant amplitude will be reduced to 50 mm when the inherent damping ratio is 0.975 %, which indicates that the equivalent damping ratio contributed by the MTMD should be higher than 0.975 % to meet the requirement of the VIV mitigation.

5.2. The uncertainty in the VIV

According to Eq. (12), the inherent damping ratio of the bridge directly affects the failure probability, and it plays the same role in the failure probability as the equivalent damping ratio. Thus, only the negative damping ratio induced by the vortex shedding is adopted for the analysis of the uncertainty in the VIV. Table 3 presents the coefficient of variation (COV) of random variables in the negative damping ratio. Fig. 6 shows the mean and COV of the negative damping ratio varying with the sample number. The COVs of random variables are their maximum value. The MCS result is stable enough when the sample number increases to 10^5, thus the sample number is determined to be 10^5. Fig. 7 shows the frequency histogram of the negative damping ratio. It can be see that the negative damping ratio is asymmetrically distributed with a mean of 1.01 % and a COV of 27.6 %, and it has non-negligible probability for the value range from 1.5 % to 2.0 %, which indicates that the uncertainty in the negative damping ratio is considerable for the VIV.

Fig. 8 shows the mean and COV of the negative damping ratio varying with the COV of each random variable. It can be seen that the varying COVs of the random variables have almost no influence on the mean of the negative damping ratio. The COV of the negative damping ratio is basically the same as those of $S_{t,s}$, Y_1, m. The COV of the negative damping ratio is twice and more than half of those of D and L_l, respectively. Since the item of $\varepsilon\alpha_4[h]^2/(4\alpha_2D^2)$ is much smaller than 1, the uncertainties in ε, α_2, α_4, and ω_h have weak influence on the uncertainty in the negative damping ratio according to the rules of the variance of independent random variables. It can be concluded that the uncertainties in $S_{t,s}$, L_l and Y_1 characterizing the occurring condition and intensity of the VIV should be seriously considered.

5.3. The equivalent damping ratio

In order to validate the accuracy on the equivalent damping ratio given by Eq. (30), two types of MTMDs without optimization are adopted to mitigate the VIV in the bridge. Table 4 presents the design parameters of the LMD and LFD MTMDs. The approximate equivalent damping ratio is calculated according to Eq. (30), and the true equivalent damping ratio is obtained from the resonance amplitude in the time domain according to.

$$\xi_{e,\text{true}} = \left(\xi_{\text{neg}} - \xi_h\right) + \xi_{\text{iden}} \tag{41}$$

where $\xi_{e,\text{true}}$ is the true equivalent damping ratio; $\xi_{\text{neg}} = 1.01\%$ is the negative damping ratio; ξ_h is the inherent damping ratio of the bridge; ξ_{iden} is the identified damping ratio according to.

$$\xi_{\text{iden}} = \frac{1}{2\pi(N+1)}\sum_{n=0}^{N}\frac{h(nT) - h(nT+T)}{h(nT+T)} \tag{42}$$

where $h(nT)$ and $h(nT+T)$ are the amplitudes of the bridge with the MTMD at time nT and $nT+T$; T is the natural period of the bridge; $N = 30$ in this numerical example.

Table 5 presents the comparison result of the equivalent damping ratio. It is noted that the true equivalent damping ratio cannot be obtained when the approximate equivalent damping ratio is smaller than $\left(\xi_{\text{neg}} - \xi_h\right) = 0.71\%$, because the resonance amplitude will no longer decay over time when the MTMD cannot eliminate completely the VIV. Thus, the resonance amplitude calculated according to Eq. (29) is used as the comparison instead of the equivalent damping ratio when $\xi_e < 0.71\%$. The comparison result shows that the two types of equivalent damping ratios or amplitudes are very close to each other in the small-value cases, but the approximate equivalent damping ratios are smaller than the true those in the large-value cases. Fig. 9 shows the relative error between the approximate result and true result. The relative error is defined as the ratio between $\xi_e - \xi_{e,\text{true}}$ and $\xi_{e,\text{true}}$. It can be seen that the magnitude of the relative error in general increases with the equivalent damping ratio in both LMD and LFD MTMD controls, and their maximum errors are –23.2 % and −14.6 %, respectively, indicating that the accuracy on the approximate equivalent damping ratio can be acceptable to characterize the MTMD's contribution on the VIV mitigation. Besides, such an error distribution does not affect the failure probability of the MTMD control.

5.4. The uncertainty in the equivalent damping ratio

The COV ranges of random variables m, ω_h, $\omega_{t,n}$, and $\xi_{t,n}$ in the equivalent damping ratio are $0 \sim 5$ %, $0 \sim 10$ %, $0 \sim 5$ %, and $0 \sim 15$ %, respectively. Fig. 10 shows the mean and COV of the equivalent damping ratio varying with the sample number. Here, the parameters of the LMD and LFD MTMDs are $\mu_T = 0.4\%$ and $\xi_t = 7\%$. The COVs of random variables are their maximum value. The result is the same as in Fig. 6 that the mean and COV are stable enough when the sample number increases to 10^5 in both LMD and LFD MTMD controls. Fig. 11 shows the frequency histogram of the equivalent damping ratio. It can be seen that the distribution of the equivalent damping ratio behaves a two-peak feature in both LMD and LFD MTMD controls, and their means and COVs are both 0.7 % and 50 %, indicating that the uncertainty in the equivalent damping ratio is complicated and significant.

Fig. 12 shows the mean and COV of the equivalent damping ratio varying with the COV of each random variable. The uncertainty in the equivalent damping ratio varying with the COV in the LMD MTMD control is similar to that in the LFD MTMD control. The mean of the equivalent damping ratio basically does not change with the COVs of m and $\xi_{t,n}$, but it decreases with the COVs of ω_h and $\omega_{t,n}$, with a reduction ratio of 28 % when the COV is 5 %. The COV of the equivalent damping ratio increases with the COVs of m, ω_h, $\omega_{t,n}$, and $\xi_{t,n}$. Among the four variables, the uncertainty in ω_h has the greatest influence on the equivalent damping ratio, with a COV of 56.4 % when the COV of ω_h is 10 %, and the following variables are $\omega_{t,n}$, m, and $\xi_{t,n}$, with COVs of 14.7 %, 5.1 %, and 2.5 % when the input COVs are 5 %, respectively.

5.5. The failure probability

According to the flowchart in Fig. 3, the optimal parameters of the MTMD or STMD can be obtained using any optimization algorithms. In order to save running time, only the parameters $S_{t,s}$, L_l, Y_1 ω_h, and $\omega_{t,n}$ are considered as the random variables to accelerate the convergence in the optimization, and their corresponding COVs are 10 %, 10 %, 10 %, 5 % , and 2 % to characterize a moderate uncertainty. The number of the MTMD is assumed as 4, three mass ratios are considered 0.2 %, 0.4 %, and 0.6 %, and three ξ_t are considered 5 %, 7 %, and 10 %. Table 7 presents the optimal parameters of the MTMDs and STMD. Since the mass ratio is small, the optimal frequency ratio of the STMD is nearly 1. In the case of $\mu_T = 0.2\%$, the optimal $\Delta\beta_M$ of the LMD MTMD is large, and the optimal $\Delta\beta_F$ of the LFD MTMD is zero, which indicates that the MTMD has the same control result as the STMD in the extreme small mass ratio case. In the cases of $\mu_T = 0.4\%$ and $\mu_T = 0.6\%$, the optimal β_1 of the LMD MTMD increases with the total mass ratio but decreases with ξ_t, and the optimal β_1 of the LFD MTMD decreases with the total mass ratio but increases with ξ_t. The optimal $\Delta\beta_M$ of the LMD MTMD decreases with the total mass ratio but increases with ξ_t, and the optimal $\Delta\beta_F$ of the LFD MTMD increases with the total mass ratio but decreases with ξ_t. These parameter fluctuations illustrate that the frequency spacing of the MTMD mitigating the VIV in bridges increases with the total mass ratio and decreases with ξ_t.

Fig. 13 shows the comparison result between the MTMD controls and STMD control. All the random variables are considered to estimate the failure probability. The COVs of ε, D, m, α_2, α_4, and $\xi_{t,n}$ are 10 %, 2 %, 2 %, 5 %, 5 %, and 10 % to characterize a moderate uncertainty. The inherent damping ratio of the bridge ξ_h is assumed as 0.3 %, and the corresponding COV is 20 %. Since the parameters of the MTMDs are approximately the same as those of the STMD when $\mu_T = 0.2\%$, the failure probabilities of the MTMDs are the same as those of the STMD. The reason for this result is that the mass ratio of each sub-TMD is approximately 0.05 %, which is too small to play its re-tuning advantage. The failure probabilities of the LMD and LFD MTMDs are 78 % and 73 % of that of the STMD when $\mu_T = 0.4\%$ and $\xi_t = 5\%$, respectively, which indicates the MTMD has played its re-tuning advantage and the LFD MTMD is more efficient. However, the superiority of the MTMD control is greatly reduced and even lost when ξ_t increases to 7 % and 10 %. Such a result also exists in the case of $\mu_T = 0.6\%$ that the failure probabilities of the LMD and LFD MTMDs are 35 % and 32 % of that of the STMD when $\xi_t = 5\%$, respectively, and the failure probabilities of the LMD and LFD MTMDs are 53 % and 51 % of that of the STMD when $\xi_t = 7\%$, respectively, while the failure probabilities of the MTMDs are slightly smaller than that of the STMD when $\xi_t = 10\%$. The reason for this result is that, (a) both the re-tuning advantage induced by the frequency spacing and the large damping of the sub-TMD can improve the robustness in mitigating the VIV in bridges, i.e., reduce the failure probability, (b) once each large damping sub-TMD becomes a robust TMD, the re-tuning advantage will no longer be significant or exist, and (c) the large damping that determines whether the sub-TMD is a robust TMD increases with the total mass ratio. We can find from the result shown in Fig. 13 that it is unnecessary to use the MTMD for mitigating the VIV in bridges if the extreme small mass ratio design is adopted, i.e., the intensity of the VIV is very weak, but the MTMD control is more welcome than the STMD if the modest or large mass ratio design is adopted, i.e., the intensity of the VIV is modest or strong.

In order to further understand the superiority of the MTMD control, Fig. 14 shows the comparison result of the fitted probability distribution curve of the limit state function between the MTMDs and STMD. The total mass ratio in Fig. 14 is 0.6 %. It can be seen that the LMD and LMD MTMDs have a similar probability distribution curve symmetrically and centrally distributed between 0 and 0.01 when $\xi_t = 5\%$ and $\xi_t = 7\%$. The probability distribution curve of the STMD is irregularly and widely distributed between -0.005 and 0.02, -0.005 and 0.015 when $\xi_t = 5\%$ and $\xi_t = 7\%$, respectively. That is to say, the STMD has an approximately equal probability for the good control, modest control, and bad control, while the LMD and LMD MTMDs have a high probability for the modest control. In fact, since the vortex-induced resonance in bridges is a self-excited and self-limited vibration, it has been completely eliminated when the limit state function is positive. Therefore, the MTMD control with good robustness is a better choice than the STMD control for mitigating the VIV in bridges. The probability distribution curve of the STMD is basically similar to those of the LMD and LMD MTMDs, with a symmetrical and central distribution, when $\xi_t = 10\%$. Since the large damping STMD is considered as a robust TMD, such a probability distribution curve shown in Fig. 14(c) validates again that the MTMD is more robust than the TMD.

6. Conclusions

This paper proposes a framework of the parameter design of the MTMD mitigating the VIV in bridges. Based on the nonlinear semi-empirical VIV model, the failure probability for the VIV is derived considering the uncertainties in the bridge and aerodynamic parameters. Furthermore, the failure probability for the VIV in bridges equipped with the MTMD is proposed as the objective function of the MTMD control. The flowchart of the parameter optimization of the MTMD is presented by simultaneously considering the failure probability and stroke limitation. A numerical example of a box girder continuous bridge is presented, including the uncertainty analysis, failure probability analysis, and comparison analysis. The conclusions that can be drawn based on the simulation results are as follows,

(1) Among all the random variables, the uncertainties in the aerodynamic parameters $S_{t,s}$, L_l and Y_1 characterizing the occurring condition and intensity of the VIV have great influence on the negative damping ratio induced by the VIV, while the uncertainties in the natural frequency of the controlled mode of the bridge and the natural frequencies of the MTMD have great influence on the equivalent damping ratio contributed by the MTMD.
(2) The optimized frequency spacing of the MTMD increases with the total mass ratio, and decreases with the damping of the MTMD. It is unnecessary to use the MTMD for mitigating the VIV with weak intensity, but the MTMD is more welcome than the STMD if the intensity of the VIV is modest or strong due to its good robustness.
(3) The failure probability of the MTMD is 30 %~50 % of that of the STMD when the total mass ratio is 0.6 %. The STMD has an approximately equal probability for the good, modest, and bad control results, while the MTMD has a high probability for the modest control result.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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